

MASS-ENG-005

Event Bus Specification

Page 1 – Purpose & Scope

Purpose

Define the enterprise messaging backbone that enables asynchronous communication between MASS HQ subsystems.

Objectives

- Decouple platform services
- Standardize event publication and subscription
- Support reliable enterprise messaging
- Enable future scalability

Out of Scope

Business workflow logic and direct application processing.

Page 2 – Architecture

Core Components

Event Publisher, Event Broker Interface, Subscription Manager, Event Catalog, Retry Handler, Dead Letter Queue Interface.

Relationships

All engines publish and consume events through the Event Bus; direct engine-to-engine dependencies should be minimized.

Page 3 – Functional Specification

Requirements

1. Support event publication.
2. Support event subscription.
3. Preserve event metadata.
4. Handle delivery failures through retry policies.
5. Route unrecoverable messages to a dead-letter queue.

Non-Functional

Reliable delivery, extensibility, observability, idempotent processing, versioned event contracts.

Page 4 – Interfaces & Examples

Interfaces

Publish(Event), Subscribe(EventType), Acknowledge(), Retry(), DeadLetter().

Example Flow

Identity Engine publishes UserCreated → Event Bus validates and routes → Notification Engine and Workflow Engine consume independently → Processing results recorded.

Page 5 – Construction Package

Claude Checklist

Implement event contract model; publisher API; subscriber API; routing layer; retry mechanism; dead-letter interface; event logging; integration tests.

Definition of Done

Modules exchange events exclusively through the Event Bus with traceable delivery and failure handling.

Constitution References

Enterprise Structure, Platform Principles, Security, Knowledge.